

Out of the Hyperintensional Bungalow: An Alternative Truthmaker Semantics for Permission

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Abstract

Fine (2014) presents a truthmaker semantics for permission. I raise some challenges for Fine's account and propose an alternative that overcomes them. The resulting framework combines alternative semantics with truthmaker semantics: the semantic value of a statement is a set of propositions (as in alternative semantics), where each proposition is a set of truthmakers (as in truthmaker semantics). Thus where alternative and truthmaker semantics are each hyperintensional 'bungalows', having only one level of hyperintensional content, I argue that permission calls for two levels of hyperintensionality.

Within philosophical logic and formal semantics, it is often thought that one can express the meaning of modal statements within possible worlds semantics. On this view, the meaning of a statement is determined by the possible worlds in which it is true (Lewis 1970, Partee 1989). According to the standard theory of modality, for example, a statement like 'Eve may sing' is true just in case there is some world in the relevant domain where Eve sings (Hanson 1965, Kratzer 1977).¹ This theory satisfies the following principle, where $\Diamond$ represents 'it is permitted that'.²

Substitution. If A and B are necessarily equivalent (true in all the same worlds), then so are $\Diamond A$ and $\Diamond B$.

Kit Fine (2014) presents a challenge to Substitution. Eve finds herself in Alternative Eden, with the forbidden apple a_0 in addition to infinitely many apples $a_1, a_2, \dots$. God tells Eve:

¹Portner (2009:47) and Chrisman (2015:69–73) call this the 'standard' theory of deontic modals.

²Following standard practice, I assume that a wide variety of constructions express the permission operator, including 'it is allowed that A ', and, where A is of the form Fa , ' a may F ' and ' a is permitted to F '.

- (1) You may eat infinitely many of the apples $a_1, a_2, a_3, \dots$

Intuitively, this does not permit Eve to eat the Forbidden Fruit a_0 . Suppose God had instead said:

- (2) You may eat infinitely many of the apples $a_0, a_1, a_2, a_3, \dots$

In contrast, this intuitively does permit Eve to eat a_0 alongside infinitely many apples.³

The puzzle is that the embedded statements are necessarily equivalent.

- (3) a. Eve eats infinitely many of the apples $a_1, a_2, a_3, \dots$
 b. Eve eats infinitely many of the apples $a_0, a_1, a_2, a_3, \dots$

If one is true, the other must be too. For a difference of one cannot separate the finite from the infinite. Possible worlds semantics of permission therefore predict God's statements in (1) and (2) to be equivalent.

Fine uses this example to motivate a truthmaker semantics of permission. He proposes that, where A is a statement, ' A is permitted' holds just in case each of A 's truthmakers is permitted. For while (3a) and (3b) are true in the same worlds, they have different truthmakers; in particular, the state of Eve eating the Forbidden Fruit a_0 alongside infinitely many apples verifies the latter but not the former. Given this, Fine accounts for the contrast between God's statements in Alternative Eden.

This paper presents three challenges for Fine's account. This first is that it misdiagnoses the status of the permission statements in Alternative Eden. Fine's account predicts that for 'Eve may eat infinitely many of the apples' to be true, Eve must be permitted to eat *any* infinite collection of the apples. This is too strong, as Fine (2014:333) himself acknowledges. Suppose, say, that Eve is permitted to every odd-numbered apple. Is she allowed to eat infinitely many of the apples? Intuitively, she is. To account for this, we argue that the inference from ' A is permitted' to 'Every truthmaker for A is permitted' is not a semantic entailment—as it is on Fine's theory—but a pragmatic inference. The contrast between God's permission statements in Alternative Eden is not

³Note that the contrast does not depend on this particular formulation of the example. As Fine (2014:329) observes, the contrast persists under a wide range of formulations; for example, suppose that the Forbidden Fruit a_0 is red, while the safe apples $a_1, a_2, \dots$ are green. Compare:

- (i) a. You may eat infinitely many of the green apples.
 b. You may eat infinitely many of the apples.

We observe the same contrast: only the second permits Eve to eat the red apple alongside infinitely many green apples.

that (1) is true and (2) is false, but that they trigger different pragmatic inferences.

The second challenge is that Fine’s account is not fine enough: it renders equivalent permission statements that permission intuitively distinguishes. To illustrate, suppose Adam may pick at most five apples from the Tree of Life. Is he permitted to pick more than four apples? Intuitively, he is, since he may pick exactly five. Is he permitted to pick five or more? Here we do not want to say that he is. Saying so would imply that he may pick more than five apples, which is forbidden. That is, the following permission statements are intuitively not equivalent.

- (4) a. Adam may pick more than four apples from the Tree of Life.
- b. Adam may pick five or more apples from the Tree of Life.

The embedded statements are true in all the same worlds: Adam picks more than four apples just in case he picks five or more. But what’s more, they arguably also have the same truthmakers—the same states are ‘wholly relevant’ to their truth. Plausibly, for each, their truthmakers are the states of Adam taking n apples, for $n = 5, 6, 7, \dots$. Given this, Fine’s truthmaker semantics of permission is not fine enough to distinguish them.

The third challenge for Fine’s account concerns its treatment of free choice, the inference from $\Diamond(A \vee B)$ to $\Diamond A$ and $\Diamond B$. We argue that it makes incorrect predictions for permission statements embedded under negation and probability operators. As it stands, the account fails to predict dual prohibition, the inference from $\neg\Diamond(A \vee B)$ to $\neg\Diamond A$ and $\neg\Diamond B$, as in the inference from ‘Eve may not have a banana or a pear’ to ‘Eve may not have a banana’ and ‘Eve may not have a pear’. Furthermore, we show that permission statements under probability operators are ambiguous between a free-choice and non-free-choice reading, while Fine’s account only predicts the free-choice reading.

Despite these challenges, we argue that Fine’s overarching claim is correct: permission requires a hyperintensional theory of semantic content. The goal of this paper is to provide such an account while overcoming the three challenges facing Fine’s theory.

Our solution combines two semantic frameworks that are usually discussed in isolation: alternative semantics and truthmaker semantics. In alternative semantics (Kratzer and Shimoyama 2002, Alonso-Ovalle 2006), the meaning of a statement is a set of propositions, called its *alternatives*, where each proposition is taken to be a set of possible worlds. Our proposal is to replace sets of worlds with sets of truthmakers. We call the resulting framework *alternative truthmaker semantics*. Thus, where alternative semantics and truthmaker semantics are ‘bungalows’, with one level of hypertensional content (alternatives or truthmakers in isolation), we argue that we need two. This

framework inherits the best of both: from truthmaker semantics we take the ability to distinguish between Eve pair in (1) and (2); from alternative semantics we take its treatment of disjunction, allowing us to distinguish between the Adam pair in (4).

We then propose that ‘ A is permitted’ is true iff for each of A ’s alternatives has at least one permitted truthmaker, and false iff each of A ’s alternatives has no permitted truthmaker. This ensures both free choice and dual prohibition, and accounts for the Adam pair. In addition, we propose that when context allows, permission statements trigger a pragmatic inference that for each alternative, *each* of its truthmakers is permitted. This will account for the Eve pair in Alternative Eden.

1 Fine’s proposal

Fine (2014) aims to account for God’s statements in Alternative Eden within truthmaker semantics (Fine 2017). Formally, we assume a propositional language $\mathcal{L}$ built from atomic sentences $At = \{p, q, \dots\}$, negation ($\neg$), conjunction ($\wedge$), and disjunction ($\vee$). Following Fine, we will admit infinitary conjunctions $A \wedge B \wedge \dots$ and disjunctions $A \vee B \vee \dots$, which will allow us to analyse *infinitely many* within a propositional language.⁴

$$A ::= p \mid \neg A \mid A \wedge B \wedge \dots \mid A \vee B \vee \dots$$

We assume the bilateral truthmaker semantics from Fine (2017), with the addition of a designated subset of the states C , what Fine (2018) calls a *code of conduct*.⁵ Our models thus have the form $(S, \sqsubseteq, V^+, V^-, C)$, where

- S is a nonempty set of states;
- $\sqsubseteq$ is a partial order that is complete: every nonempty set of states T has a least upper bound, called its *fusion*, denoted $\bigsqcup T$;
- $V^+ : At \rightarrow \mathcal{P}(S)$ is an exact verification relation, determining for each atomic sentence the set of its exact verifiers;
- $V^- : At \rightarrow \mathcal{P}(S)$ is an exact falsification relation, determining for each atomic sentence the set of its exact falsifiers;

⁴Those who object to an infinitary language (see Barwise 1981) may instead analyse *infinitely many* F s are G within a suitably enriched first-order language as $\exists I \subseteq^\infty F : \forall x \in I : Eat(Eve, x)$, where $X \subseteq^\infty Y$ denotes that X is an infinite subset of Y ; see Fine (2017:566–69) for a first-order truthmaker semantics. However, providing a truthmaker semantics for first-order logic raises special difficulties; principally, regarding whether the domain can vary from world to world, and whether the truthmakers for universally quantified sentences include totality facts. Since these issues do not impact our remarks here, we follow Fine in analysing the sentence via infinitary disjunction and conjunction.

⁵The differences between the bilateral and unilateral semantics will not concern us here: the remarks to follow apply to the unilateral and bilateral versions equally.

$C \subseteq S$ is a set of states; intuitively, those that are permitted.

The verification and falsification relations V^+ and V^- is assumed to be exact, in the sense that whenever a state verifies (respectively, falsifies) a statement, the state's obtaining is 'wholly relevant' to the sentence's truth (falsity). We define for each statement A the set of its exact verifiers (denoted $|A|_M^+$) and exact falsifiers (denoted $|A|_M^-$) with respect to a model M , adopting the standard clauses from bilateral truthmaker semantics, generalised to allow infinitary conjunction and disjunction.⁶

$$\begin{aligned}
|p|^+ &= V^+(p) \\
|p|^- &= V^-(p) \\
|\neg A|^+ &= |A|^- \\
|\neg A|^- &= |A|^+ \\
|A \wedge B \wedge \dots|^+ &= \{s \sqcup t \sqcup \dots : s \in |A|^+, t \in |B|^+, \dots\} \\
|A \wedge B \wedge \dots|^- &= |A|^- \cup |B|^- \cup \dots \\
|A \vee B \vee \dots|^+ &= |A|^+ \cup |B|^+ \cup \dots \\
|A \vee B \vee \dots|^- &= \{s \sqcup t \sqcup \dots : s \in |A|^-, t \in |B|^- , \dots\}
\end{aligned}$$

Fine assumes the natural model of the Alternative Eden scenario. For each apple a , the state space contains the state s_a of Eve eating apple a , as well as the fusions thereof. Each state s_a exactly verifies the atomic sentence p_a that Eve eats a . He then regiments the Eve pair as follows:

- (5) a. Eve eats infinitely many of $a_1, a_2, \dots$ $\bigvee_{A \subseteq \infty \{a_1, a_2, \dots\}} \bigwedge_{a \in A} p_a$
 b. Eve eats infinitely many of $a_0, a_1, a_2, \dots$ $\bigvee_{A \subseteq \infty \{a_0, a_1, a_2, \dots\}} \bigwedge_{a \in A} p_a$

These are true in all the same worlds but have different verifiers.⁷ In particular, the state of Eve eating a_0 alongside infinitely many of the apples from Alternative Eden

⁶In what follows, we omit the subscript M to reduce notational clutter. The infinitary clauses may be stated more formally as follows, where the i 's are drawn from some (perhaps infinite) index set I : $|\bigwedge_i A_i|^+ = \{\sqcup_i f(i) \mid f : I \rightarrow S, f(i) \in |A_i|^+ \text{ for all } i\}$; $|\bigvee_i A_i|^+ = \bigcup_i |A_i|^+$; $|\bigwedge_i A_i|^- = \bigcup_i |A_i|^-$; $|\bigvee_i A_i|^- = \{\sqcup_i f(i) \mid f : I \rightarrow S, f(i) \in |A_i|^- \text{ for all } i\}$.

⁷One may worry that (3a) and (3b) are not necessarily equivalent for the following reason. Take a world where a_0 does not exist and Eve eats infinitely many of $a_1, a_2, a_3, \dots$. In such a world, (3a) is clearly true, but one might think (3b)'s truth value is undefined due to presupposition failure, owing to the non-referring term a_0 , in the same way one might take 'the King of France is not bald' to be undefined (Strawson 1950). In response, consider sentences explicitly affirming that the apples $a_0, a_1, a_2, \dots$ exist:

- (i) a. There are infinitely many apples $a_0, a_1, \dots$, and you may eat infinitely many of $a_1, a_2, \dots$.
 b. There are infinitely many apples $a_0, a_1, \dots$, and you may eat infinitely many of $a_0, a_1, a_2, \dots$.

verifies the latter but not the former.

In Alternative Eden, it is clear that every state of Eve eating infinitely many of the safe apples $a_1, a_2, \dots$ is permitted (for any $T \subseteq \{s_{a_1}, s_{a_2}, \dots\}, \sqcup T \in C$), while any state containing the state of Eve eating the Forbidden Fruit a_0 is not (for any state t such that $s_{a_0} \sqsubseteq t, t \notin C$).

He then proposes the following semantics of permission statements (Fine 2014:335).⁸

(6) $\Diamond A$ is true iff all of the verifiers of A are permitted.

This correctly predicts that in the context of Alternative Eden, (1) is true but (2) is not—the desired result.

Fine is careful to distinguish between the descriptive and prescriptive uses of permission statements. He intends the clause in (6) to represent the descriptive use of permission statements: what it means for one to truly describe the existing permissions. Though as Fine makes clear, God’s pronouncements in Alternative Eden are no mere descriptions of what Eve may do. Rather, they *create* permissions. For this reason, Fine also proposes a corresponding clause for the prescriptive use of permission statements.⁹

(7) $\Diamond A$ permits all and only those actions which verify A .

For the same reason as the descriptive use, this clause successfully distinguishes between the prescriptive use of (1) and of (2). Only the latter permits Eve to eat the Forbidden Fruit a_0 along with infinitely many apples.

Beyond accounting for the Alternative Eden case, Fine’s theory has further desirable properties. One notable feature, discussed by Fine (2018:§6), is that it accounts

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- (ii) a. You may look at every apple $a_0, a_1, \dots$ and eat infinitely many of $a_1, a_2, \dots$.
 - b. You may look at every apple $a_0, a_1, \dots$ and eat infinitely many of $a_0, a_1, a_2, \dots$.

The contrast persists: only the (b)-sentences permit Eve to eat a_0 alongside infinitely many apples.

⁸Fine later refined his theory of permission since (Fine 2018), proposing that X is permitted just in case “every course of action in compliance with X is contained in an ideal action” (Fine 2018:639). He also introduces the notion of weak permission, whereby X is weakly permitted iff every action in compliance with X is compatible with a course of action sanctioned by the code (Fine 2018:642). These clauses inherit universal quantification over verifiers—our main target in what follows—so the challenges we present below apply equally to these updated clauses. For simplicity, in what follows we stick with the simpler clause in (6).

⁹For discussion of the difference between descriptive and prescriptive uses of permission statements, see Austin (1962), Kamp (1978), Lewis (1979), van Rooij (2000), and Kaufmann (2012).

for free choice, the inference from $\Diamond(A \vee B)$ to $\Diamond A$ and $\Diamond B$ (Ross 1941, Kamp 1973, Merin 1992). Fine later appeals to Ross’s problem of free choice in support of universal realisability: “suppose X is permissible. Then it is plausible (at least for those of us who have Ross-type intuitions!) that every action in compliance with X is permissible” (Fine 2018:640).

Free choice is intuitively valid on both the descriptive and prescriptive uses of permission statements. On the descriptive use, if it is true that one is permitted to eat ice cream or cake, $\Diamond(A \vee B)$, it intuitively follows that they are permitted to eat ice cream and that they are permitted to eat cake, $\Diamond A \wedge \Diamond B$. And on the prescriptive use, if one is informed that they may eat ice cream or cake (by someone with the relevant authority, such as a parent to a child), it intuitively follows that they are permitted to eat ice cream and permitted to eat cake. These inferences fall out automatically on Fine’s account, given the clauses in (6) and (7), and that a state verifies a disjunction just in case it verifies one of the disjuncts.

Fine’s semantics achieves its goal of distinguishing between (1) and (2). This is a noteworthy achievement, given that the embedded statements are necessarily equivalent, and therefore indistinguishable within possible worlds semantics of permission. There are, however, a number of challenges facing Fine’s account.

2 Challenge 1: Coarse permission

Fine’s semantics is designed to satisfy the following principle, where we take the ‘ways’ for a statement to obtain to be its verifiers.¹⁰

(8) **Universal Realisability of Permission.**

A statement is permitted just in case every way for it to obtain is permitted.

This marks a radical departure from the standard semantics of permission, according to which $\Diamond A$ is true just in case A is true in some deontically ideal world. Thus on the standard semantics, permission is *coarse*: a statement can be permitted even if some ways for it to obtain are not permitted.

2.1 Evidence for coarseness

Is permission coarse? Consider:

¹⁰This is in analogy with Fine’s truthmaker semantics for counterfactuals, which satisfies a principle Fine calls *Universal Realisability of the Antecedent*: a counterfactual is true when it is true for any way in which its antecedent might be true (Fine 2012:236).

- (9)
- a. There are two switches, A and B. Each switch can be either up or down. The rules specify that A must be up and that B may be in any position. Are the positions of the switches allowed to agree?
 - b. Alice and Bob are colleagues in Quebec. They know both English and French. Bill 96 requires them to speak French at work, but they defy the law by talking to each other at work in English. Does the new rule permit Alice and Bob to talk to each other at work?
 - c. John's doctor has permitted him to smoke at most seven cigarettes per day. Does John have permission to smoke more than six cigarettes per day?
 - d. Alice has many tins of paint in various shades of red, yellow, blue, and so on. Bob asks Alice to paint his house. He chooses three paints he likes, called Scarlet, Sunflower Yellow, and Yves Klein Blue. He cannot decide between them, so he asks Alice to paint his house using whichever of the three paints she prefers. Is Alice allowed to paint Bob's house red?

Intuitively, the answer to each is 'Yes'. For example, Alice and Bob are allowed to talk to each other at work, since they may do so in French, even though they may not do so in English. John is permitted to smoke more than six cigarettes a day, since he may smoke seven.¹¹

It is plausible that the state of the switches being both down exactly verifies that they agree, that the state of Alice and Bob talking to each other at work in English exactly verifies that they talk to each other at work in French, and so on for the other examples. Given this, the judgements above are evidence for existential realisability, that a permission statement $\Diamond A$ is literally true iff *some* verifier of A is permitted.

We find similar judgements in Alternative Eden. Consider:

- (10) God told Eve that she may eat every odd-numbered apple. Is Eve permitted to eat infinitely many of the apples?

Intuitively, she is. (1) and (2), on their descriptive use, are both intuitively true in this scenario. This is evidence that existential realisability is also correct for Fine's example.

Fine acknowledges something to this effect:

¹¹This example comes from Fox (2007:103), who reports the same judgement.

There are also naturally occurring examples of coarseness, such as: "the injured party is allowed to be present during investigative acts in the phase of a judicial investigation. Specifically, he is allowed to be present at [...] the questioning of a witness, but only if it is expected that the witness will not come to the trial" (Zgaga 2011:89); "Am I allowed to have help when I am voting? Yes! In many forms" (Maine Students Vote 2025). Being present during investigative acts is allowed, but not in all cases. You are allowed to have help when voting, in many forms (but presumably not all).

In the particular case of God's permitting Eve to eat infinitely many apples, we may obtain a perfectly good interpretation of the statement by taking it to be true just in case, for some infinite number of the apples, God has in fact granted Eve permission to eat those apples.

(Fine 2014:333)

However, Fine's own theory does not respect his observation. It instead predicts the statement 'Eve may eat infinitely many of the apples' to be true just in case, for *every* infinite collection of apples, God has granted Eve permission to eat those apples. Thus at the heart of Fine's paper lies an important tension: Fine acknowledges that existential realisability offers 'a perfectly good interpretation' of the permission statements in question, but his clauses enforce universal realisability.

2.2 Objections and replies

Before moving on, let us address two objections one may raise against this argument for existential realisability.

Contextual restriction. One might take a broad view of verification, on which a state's verifiers go far beyond those that are relevant or salient in a particular conversation. Given this, Universal Realisability may seem hopelessly strong. Fine offers the examples: 'One is allowed to talk in normal social situations, but not too loudly, and to approach other people, but not too closely' (Fine 2014:320). Plausibly, the state of talking loudly verifies that one is talking, and the state of approaching within 1cm of another person verifies that one approaches them. These are counterexamples to Universal Realisability provided that the state of talking loudly verifies that one is talking, and the state of approaching within 1cm of another person verifies that one approaches them. This suggests that we should restrict Universal Realisability (on both the descriptive and prescriptive uses) to a set of contextually relevant verifiers.

While some form of context-dependence is likely required by Fine's account, I doubt that it overcomes the problem with (9). Recall (9a). Both ways for the switches to be in the same position appear to be on a par in terms of their contextual relevance. The situation is even more stark with (9b). Recall that 'Adam and Bob are allowed to talk to each other at work' is intuitively true in a scenario where they are allowed to talk to each other in French, even though they actually violate the rule by talking to each other in English. This fails the Universal Realisability test since Adam and Bob talking together in English is forbidden. For contextual restriction to overcome the problem, we would have to say that Adam and Bob talking together in English—the actual way

in which they do so—is contextually irrelevant. Plausibly, however, if any verifiers of a statement are contextually relevant, its actual verifiers are.

Scope ambiguity. One might reply that ‘Eve may eat infinitely many of the apples’ exhibits a scope ambiguity, with $\Diamond$ scoping either above or below *infinitely many*, as $\Diamond \exists A' \subseteq^\infty A : Ex$ (‘It is permitted that Eve eat infinitely many of the apples’) or $\exists A' \subseteq^\infty A : \Diamond Ex$ (‘For infinitely many apples, Eve may eat them’). Indeed, other counting quantifiers such as *at most* and *at least* are known to exhibit scope flexibility with respect to permission (Geurts and Nouwen 2007, Buring 2008).

However, even if this is possible for *infinitely many*, it fails to save universal realisability from the counterexamples in (9). For instance, there is no syntactic parse of ‘Alice and Bob may talk to each other’ as ‘There is a language L such that Alice and Bob may talk to each other in L ’, nor a parse of ‘The positions of the switches are allowed to agree’ as ‘There is a position P of the switches such that the switches are allowed to agree in’.¹²

2.3 Universal realisability as a pragmatic inference

While the intuitive answer to the questions in (9) is ‘Yes’, there is something misleading about asserting the permission statements in (11) outright in their respective contexts.

- (11)
- a. The switches are allowed to agree.
 - b. Adam and Bob are allowed to talk to each other at work.
 - c. John has permission to smoke more than six cigarettes per day.
 - d. Adam is allowed to paint Bob’s house red.
 - e. Eve is permitted to eat infinitely many of the apples.

This is especially clear if these statements are not given in response to a Yes/No question, asking whether or not the statement is true, but in response to an open-ended question about what is permitted. In the case of (11a), for example, if one is asked ‘What positions of the switches are allowed?’ or ‘Tell me about the rules governing the switches’, the answer ‘The switches are allowed to be in the same position’ seems deeply misleading. It naturally suggests that *every* way for the switches to be in the same position is allowed; that they may both be up and that they may both be down. The same goes for the other examples in (11), which in response to such questions have the inferences in (12), respectively.

¹²Though note that existential realisability predicts these to be truth-conditionally equivalent, and hence hard to tease apart semantically.

- (12)
- a. The switches are allowed to be up and they are allowed to be down.
 - b. Adam and Bob are allowed to talk to each other at work in any language.
 - c. John has permission to smoke seven, eight, nine, ... cigarettes per day.
 - d. Adam is allowed to paint Bob's house any shade of red.
 - e. For every infinite collection of apples, Eve is permitted to eat every apple in the collection.

Let us call these *universal realisation inferences*.

(13) **Universal Realisation Inference.**

The inference from $\Diamond A$ to: every truthmaker for A is permitted.

Universal realisation inferences are best thought of as implicatures, derived by pragmatic reasoning. Here we show this using two tests for pragmatic inference: question-sensitivity and cancellability.

Test 1. Question-sensitivity. Pragmatic inferences are known to be highly sensitive to the question under discussion (Welker 1994, von Kuppevelt 1996, Breheny, Katsos, and Williams 2006, Zondervan 2009, Degen and Tanenhaus 2015, Ronai and Xiang 2021). A classic example is the inference from *some* to *not all*. If Adam talked to all of the students, and Bob asks, 'Did Adam talk to some of the students?', it is natural to answer 'Yes'. If Bob had instead asked 'How many students did Adam talk to?', the answer 'Some of them' strongly suggests that she did not talk to all of them. As we have seen, universal realisation inferences behave in the same way. In response to the polar question 'Is A permitted?', the natural answer in (11) is 'Yes', while in response to 'Under what conditions is so-and-so permitted?' or 'Tell me about the rules governing so-and-so', responding with ' A is permitted' strongly suggests that every way to do so is permitted.

Test 2. Cancellability. Universal realisation inferences are cancellable. It is consistent to say, 'The switches are allowed to agree. More specifically, they are allowed to be up both, but they are not allowed to be both down'. If the literal truth conditions of permission statements entailed the universal realisation inferences, we would expect this to be equivalent to: 'Every way for the switches to agree is allowed. More specifically, they are allowed to be up both, but they are not allowed to be both down', which is inconsistent.

Similarly, it is consistent to say, 'Eve may eat infinitely many of the apples. More specifically, she may only eat the odd-numbered apples'. But the following is inconsis-

tent: ‘For each infinite collection of apples, Eve may eat every apple in the collection. More specifically, she may only eat the odd-numbered apples’.

2.4 Coarse vs fine: descriptive vs prescriptive

So far in discussing universal realisation inferences, we have ignored the distinction between descriptive and prescriptive uses of permission statements. It is worth noting that the universal realisation inferences arise on *both* uses. For the descriptive use, we may imagine one who is informed about the relevant rules but lacks the authority to make them. When such a speaker asserts (11), we still tend to draw the respective universal realisation inferences in (12), especially in response to a question about what is permitted in general (such as ‘What positions of the switches are allowed?’).

This shows that universal realisation inferences are distinct from the effects of deontic update, which have also been treated within truthmaker semantics (Yablo 2011, Fine 2018, Rothschild and Yablo 2023). In cases of deontic update, one with the relevant authority *changes* what is permitted, while universal realisation inferences can arise even for permission statements made by an independent observer with no power to change what is permitted.

If we instead imagine a speaker with the authority to make the rules, such as the God of Alternative Eden, using the permission statement prescriptively, the universal inferences are especially strong. Plausibly, this is because the prescriptive use tends to answer an open-ended question such as ‘What is permitted?’ rather than the polar question ‘Is A permitted?’. As we have seen, answers to the former, what-is-permitted question invite universal realisation inferences much more strongly than answers to the latter, is-this-permitted kind.

2.5 Interim summary

Fine’s theory of permission misdiagnoses the status of the inferences in Alternative Eden. The statement ‘Eve may eat infinitely many of the apples $a_0, a_1, \dots$ ’ is literally true when *some* verifier of ‘Eve eats infinitely many of the apples $a_0, a_1, \dots$ ’ is permitted. Depending on the context (more specifically, the question under discussion), the permission statement may carry the implicature that *every* verifier of the statement ‘Eve eats infinitely many of the apples $a_0, a_1, \dots$ ’ is permitted. Fine’s account, as it stands, represents this universal realisation inference as an entailment, and therefore does not respect the coarseness of permission, nor Fine’s own remarks about when a permission statement is true.

3 Challenge 2: Fine is not fine enough

The second challenge for Fine’s account is that it is not fine enough: it renders equivalent sentences that are intuitively distinct. Consider:

- (14) *Adam may pick at most five apples from the Tree of Life.*
- a. Adam may pick more than four apples from the Tree of Life.
 - b. Adam may pick five or more apples from the Tree of Life.

Intuitively, Adam may pick more than four apples, since he may pick five. But it is wrong to say that he may pick five or more, since he may not pick more than five. Now, the embedded statements are necessarily equivalent: Adam picks more than four apples if and only if he picks five or more. But more than that, it is plausible that they are also *exactly* equivalent, that they have the same exact verifiers.

The guiding idea behind truthmaker semantics is that a statement’s verifiers are ‘wholly relevant’ to its truth (Fine 2017). Let us then ask what states are wholly relevant to the truth of the embedded statements:

- (15) a. Adam picks more than four apples.
b. Adam picks five or more apples.

Plausibly, for both, they are the state of Adam picking n apples, for $n = 5, 6, 7, \dots$. Assuming that the embedded sentences in (15) are exactly equivalent, Fine predicts the permission statements in (14) to also be equivalent—the wrong result.

One might doubt this assumption on exact verification. After all, it is not forced on Fine’s theory. Exact verification is a primitive of the account, so we are free, within the constraints of the theory, to set the exact verification relation as best fits our judgements.

One might therefore consider introducing an indeterminate state of Adam eating more than four apples, one exactly verifying that he picks more than four but not precisely how many he picks.¹³ We may then suppose that (15b) has such indeterminate

¹³Note that the addition of indeterminate states is not altogether innocent. It clashes with Fine’s (2011) treatment of the determinate/determinable distinction, where he requires that all states be fully determinate; specifically, see Fine (2011:168) on Directed Completeness. We would also lose the exact entailment from ‘the ball is scarlet’ to ‘the ball is red’. Moreover, it is extremely difficult to say what mereological relations indeterminate states should stand in. Plausibly, the state of the ball being red is a proper part of the state of it being scarlet. Classical mereological principles would then imply the existence of a remainder—the part of the scarlet state disjoint from the red state—giving determinates a *species = genus + differentia* structure, e.g. scarlet = red + remainder, they are argued to lack (Johnson 1921, Funkhouser 2006, Wilson 2023).

states as verifiers. For disjunction, exact verification is constrained by the semantic clause as before: a state exactly verifies a disjunction iff it exactly verifies some disjunct. Then the verifiers of (15b) would be states of Adam eating exactly five apples, as well as indeterminate states of him eating more than five apples. On this view, then, (15a) and (15b) would no longer have the same verifiers.

There are, however, other cases of exact equivalence that cannot be handled by modifying the exact verification relation in this way. The truthmaker clauses guarantee De Morgan's law, that $\neg(A \wedge B)$ and $\neg A \vee \neg B$ have the same verifiers. But these come apart under deontic modals:

- (16) a. You have to work Saturday, but you don't have to work both Saturday and Sunday. $\Box A \wedge \neg \Box(A \wedge B)$
b. You have to work Saturday, but you may take off either Saturday or Sunday. $\Box A \wedge \Diamond(\neg A \vee \neg B)$

The first is fine, but the second sounds contradictory. Permission to take off either Saturday or Sunday implies permission to take off Saturday, contradicting the obligation to work Saturday.¹⁴

This example concerns obligation, whereas we have so far been solely concerned with permission. But we can construct analogous examples purely in terms of permission, as in (17).¹⁵

- (17) a. It is not permitted to take Saturday off, but it is permitted to not work both days this weekend. $\neg \Diamond \neg A \wedge \Diamond \neg(A \wedge B)$
b. It is not permitted to take Saturday off, but it is permitted to take either day off this weekend. $\neg \Diamond \neg A \wedge \Diamond(\neg A \vee \neg B)$

These elicit the same contrast: the first is fine but the second sounds contradictory.¹⁶ This is a problem for theories of permission that validate De Morgan's Law, including Fine (2014, 2018), Aloni (2018), and Willer (2018).

These counterexamples go beyond the question of whether existential or universal

¹⁴The inference from $\neg \Box(A \wedge B)$ to $\neg \Box A \wedge \neg \Box B$ is known as *negative free choice*. Marty et al. (2021) found that while the inference is sometimes drawn, it is far less robust than the free choice inference from $\Diamond(A \vee B)$ to $\Diamond A \wedge \Diamond B$. To account for this, they propose that free choice is an entailment but negative free choice is an implicature.

¹⁵Indeed, (16) itself is not a problem for Fine (2018:649), who rejects that $\neg \Box A$ implies $\Diamond \neg A$. (17) is nonetheless a problem for Fine.

¹⁶Note that the phrase 'it is permitted to not work both days' exhibits a scope ambiguity, with available readings $\Diamond \neg(A \wedge B)$, $\Diamond(\neg A \wedge \neg B)$ and $\Diamond \neg A \wedge \Diamond \neg B$. We are interested in the first reading.

realisability is correct. Universal and existential realisability both give wrong results here. Universal realisability— $\Diamond A$ is true iff every verifier of A is permitted—makes (14a) and (17a) too strong, predicting them to be false. Existential realisability— $\Diamond A$ is true iff some verifier of A is permitted—makes (14b) and (17b) too weak, predicting them to be true.

Prior observed this tension back in 1959.

“All ways of doing α or β ” is indeed equivalent to “All ways of doing α and all of doing β ”; but we would not ordinarily infer “It is permitted either to smoke or to embezzle” from “It is permitted to smoke.” [...] but α (say my feeding this man) is usually considered permissible even if some ways of doing α (e.g. feeding him with poison) are not, so long as some are; and smoking is one way of smoking-or-embezzling.

(Prior 1959:178)

The lesson is that universal realisability holds for permissions of disjunctions, while existential realisability holds for permissions of non-disjunctive sentences. That is, free choice is valid: $\Diamond(A \vee B)$ implies $\Diamond A$ and $\Diamond B$ (Ross 1941, Kamp 1973, Merin 1992), but this should not lead us to conclude that for any statement A whatsoever, $\Diamond A$ is true iff all the ways for A to hold (in truthmaker semantics speak, its verifiers) are permitted.

3.1 Free choice is not a universal realisation inference

It is worth pointing out that the free choice inference from $\Diamond(A \vee B)$ to $\Diamond A \wedge \Diamond B$ differs in status from universal realisation inferences discussed in Section 2.3. It fails both of the tests considered there. First, it is not sensitive to the question under discussion in the way universal realisation inferences are. The natural answer in this context to ‘Is Adam allowed to pick more than four apples?’ is ‘Yes’, while giving the same answer to ‘Is Adam allowed to pick five or more apples?’ is deeply misleading.

Second, it is often observed that free choice inferences are not cancellable (Simons 2005, Barker 2010, Starr 2016, a.o.). Compare: ‘Adam may pick more than four apples. In fact, he may pick at most five’ and ‘Adam may pick five or more apples. In fact, he may eat at most five’. There is a striking contrast: the first is acceptable but the second is not.¹⁷

¹⁷In addition, Chemla and Bott (2014) and Tieu et al. (2016) report that free choice inferences are acquired before pragmatic inferences such as scalar implicatures.

4 Challenge 3: Embedded permission

The third challenge facing Fine’s account comes from embedding permission statements under further operators. Here we focus on two environments in particular: negation and probability.

4.1 Permission under negation

It has often been observed that a sentence of the form ‘you may not A or B ’ implies both ‘you may not A ’ and ‘you may not B ’, a phenomenon known as *dual prohibition* (Alonso-Ovalle 2006, Willer 2018, Goldstein 2019, Aloni 2022). Fine observes that this raises a potential problem for his account.

It might be thought that the present proposal is subject to a difficulty with negative permission statements. For, intuitively, ‘you may not have candy or ice cream’ implies ‘you may not have candy’ and ‘you may not have ice cream’, whereas, on the present proposal, it should only imply ‘you may not have candy or you may not have ice cream’. However, it is far from clear that the ‘not’ in the first of these sentences is an external negation, governing the sentence ‘you may have candy or ice cream’. Rather, it governs the operator ‘may’ (or the permission operator P) and the resulting operator ‘may not’ (or F [for ‘Forbidden’]) is then subject to a clause similar to the one for permission; ‘ $F[A]$ ’ will be true iff all the verifiers of A are not permitted.

(Fine 2014:336)

This response may be plausible for *may not* (which is anyway peculiar: the negation appears below the modal, as $P\neg$, but is interpreted above it, as $\neg P$; see Cormack and Smith 2002, Iatridou and Zeijlstra 2013, Jeretič and Thoms 2023). However, it is much less plausible for other constructions:

- (18)
- a. You are not allowed to have candy or ice cream.
 - b. No one is allowed to insult their mother or their father.
 - c. I doubt that you have permission to take Monday or Tuesday off work.
 - d. Q: May I eat my candy or ice cream on the bus?
A: No.

These examples clearly feature a negated permission claim, and in (18b), $\neg\exists x\Diamond(Fx \vee Gx)$, the negative quantifier *no* must scope above *their* in order to bind it. And in (18c),

the negation is not expressed overtly but contributed by the meaning of *doubt*, assuming that *doubt that S* is equivalent to *think that $\neg S$* .

In each case, we observe the strong inference that *both* options are forbidden. However, Fine only predicts the weak inference that at least one of the options is forbidden.¹⁸

4.2 Permission under probability operators

Similar difficulties arise under probability operators. Consider the following scenario.

It's time for dessert. Alice's parents play a game to decide what dessert she is allowed to have. They roll a die. If it lands 1 or 2, she is allowed fruit (and only fruit). If it lands 3 or 4, she is allowed ice cream (and only ice cream). If it lands 5 or 6, she is allowed cake (and only cake). The die has just been cast, but we don't know how it landed. What is the probability that Alice is allowed to have ice cream or cake?

We can easily access a reading on which the answer is two thirds. After all, there is a two in three chance that the die lands 3, 4, 5, or 6.

The question also intuitively has a reading on which the answer is zero, brought out by prosodic focus on *or*. On this reading, the question asks for the probability that Alice may choose between ice cream or cake; that is, that she may have ice cream and that she may have cake. But there is no chance of that happening, since no matter how the die lands, she is allowed at most one dessert.

¹⁸A further point Fine offers in favour of his proposal is that 'If we employ what is explicitly an external negation, as in "it is not the case that: you may have candy or ice cream", then we get the weaker reading, as predicted by the present proposal' (Fine 2014:336). However, it is doubtful whether 'it is not the case' must express literal, sentential negation. According to Horn (1989), even constructions like 'it is not the case that' can express metalinguistic negation. This is illustrated in (i).

- (i) a. It is not the case that John ate SOME of the cookies; he ate ALL of them.
- b. It is not the case that they had a baby and got married; they got married and had a baby.
(based on Wilson 1975:151)
- c. It is not the case that I LIKE you; I LOVE you.

Metalinguistic negation can target any objectionable feature of a sentence whatsoever (such as its inferences, choice of words, or pronunciation), not merely its truth value. Thus it is entirely possible that in Fine's example, 'it is not the case that: you may have candy or ice cream', the metalinguistic negation expressed by 'it is not the case that' is rejecting the free choice inference (that you may have candy and may have ice cream). This derives the weak reading Fine observes. Tellingly, the weak reading Fine is after comes through strongly with prosodic focus on *or*—a hallmark of metalinguistic negation. Given the availability of metalinguistic negation, Fine's observation does not support the universal semantics in (6).

We are interested in the first reading (we return to the second reading in Section 5.3). On this reading, the probability of $\Diamond(A \vee B)$ is greater than that of $\Diamond A \wedge \Diamond B$. This, however, is impossible on a semantic theory of free choice such as Fine's, on which $\Diamond(A \vee B)$ entails $\Diamond A \wedge \Diamond B$. It is standardly assumed that entailment preserves probability: if A entails B , the probability of A cannot exceed that of B (see Yalcin 2010)—a consequence of Kolmogorov's axioms of probability. For example, it cannot be more probable that John is in Berlin than that he is in Germany. Similarly, if $\Diamond(A \vee B)$ entails $\Diamond A \wedge \Diamond B$, the probability of $\Diamond(A \vee B)$ cannot exceed that of $\Diamond A \wedge \Diamond B$, contrary to our judgements in this scenario.¹⁹

These challenges disappear if we replace Fine's universal clause in (6) with its existential analogue:

(19) **Existential realisability.**

$\Diamond A$ is true iff at least one of A 's verifiers is permitted.

For then $\neg\Diamond A$ will hold iff *none* of A 's verifiers are permitted. Thus $\neg\Diamond(A \vee B)$ will imply $\neg\Diamond A$ and $\neg\Diamond B$, correctly predicting the inferences in (18). And since $\Diamond(A \vee B)$ will be equivalent to $\Diamond A \vee \Diamond B$, the probability that Alice may have ice cream or cake will be the probability that she may have ice cream or she may have cake, which in the scenario above is 2/3.

The principal difficulty with this repair is that it fails to account for free choice, discussed in Section 3. Given the truthmaker clause for disjunction in ??, the fix in (19) predicts $\Diamond A$ to imply $\Diamond(A \vee B)$. It therefore fails to account for the contrast in (14). Existential realisability predicts both 'Adam may pick more than four apples' and 'Adam may pick five or more apples' to be equally true when she may eat at most five, contrary to intuition.

¹⁹One may try to resist the problem by arguing that the question 'What is the probability that Alice is allowed to have ice cream or cake?' in fact asks for the probability of $\Diamond A \vee \Diamond B$, with the disjunction taking wide scope. We may respond by rephrasing the problem. Consider (i).

- (i) Alice is allowed to have ice cream or cake.

All theories (including Fine's) agree that with (i), what you see is what you get: they parse it as $\Diamond(A \vee B)$, with permission scoping above the disjunction. Now, what is the probability that (i) is true? The answer is intuitively the same: two thirds.

5 A positive proposal

Fine’s theory of permission faces a dilemma. Coarse permission and embedding data from negation and probability support existential realisability, while free choice supports universal realisability. Fine opts for universal realisability, thereby capturing free choice but not the embedding data. Opting for existential realisability would instead leave free choice unaccounted for. How to escape the dilemma?

Our solution will be to combine two semantic frameworks that are usually are discussed in isolation: truthmaker semantics and alternative semantics (Kratzer and Shimoyama 2002, Alonso-Ovalle 2006, Aloni 2007). The resulting theory will combine the best of both. From truthmaker semantics, we take the ability to distinguish necessarily equivalent sentences; in particular, the Eve pair, ‘Eve may eat infinitely many of the apples $a_0, a_1, a_2, \dots$ ’ and ‘Eve may eat infinitely many of the apples $a_1, a_2, \dots$ ’. From alternative semantics, we take its treatment of disjunction. This will allow us to distinguish the Adam pair: ‘Adam may pick more than four apples’ and ‘Adam may eat five or more apples’. These intuitively have the same exact verifiers (the state of Adam picking n apples, for $n = 5, 6, \dots$), meaning truthmaker semantics cannot distinguish them. But as we will see, alternative semantics can.

The Eve pair and the Adam pair together illustrate an important point: hyperintensionality has many sources. Each pair consists of necessarily equivalent sentences. The fact that in each pair, the sentences exhibit different behaviour under permission calls for a hyperintensional notion of content. However, the two pairs differ in how their hyperintensionality comes about. In the Eve pair, it is due to a difference in their verifiers. In the Adam pair, it is due to the presence of disjunction.

5.1 Alternative truthmaker semantics

Possible worlds semantics takes a statement’s semantic content to be a proposition, understood as the set of possible worlds where the statement is true (Lewis 1970, Partee 1977). Alternative semantics goes one level up, taking a statement’s semantic content to be a set of propositions (Hamblin 1973, Kratzer and Shimoyama 2002, Alonso-Ovalle 2006). These propositions are called the statement’s *alternatives*. For instance, the semantic value of a disjunction $A \vee B$ is taken to be

$$\{\{w : A \text{ is true at } w\}, \{w : B \text{ is true at } w\}\},$$

with two alternatives: the set of worlds where A is true, and the set of worlds where B is true.²⁰

Our sole modification to alternative semantics will be to replace sets of worlds with sets of truthmakers. Thus, for instance, the semantic value of $A \vee B$ will now be

$$\{\{s : s \text{ exactly verifies } A\}, \{s : s \text{ exactly verifies } B\}\}.$$

Naturally, we call the resulting theory *alternative truthmaker semantics*.

Replacing sets of worlds with sets of truthmakers requires modifying the entry for conjunction. In standard alternative semantics, conjunction denotes pairwise intersection: $\llbracket A \wedge B \rrbracket = \{P \cap Q : P \in \llbracket A \rrbracket, Q \in \llbracket B \rrbracket\}$. In truthmaker semantics, conjunction is not represented by intersection—states are not sets after all, and so cannot be intersected—but by fusion. Accordingly, we take conjunction to denote pairwise fusion. Given n sets of states $P, Q, \dots$, let us abuse notation and write $P \sqcup Q \sqcup \dots = \{s \sqcup t \sqcup \dots : s \in P, t \in Q, \dots\}$ for their n -wise fusion.²¹

A further difference between truthmaker and alternative semantics is that the latter comes with a ‘flattening’ operator $!$ (sometimes denoted ‘ $\exists$ ’) whose sole function is to ‘flatten’ alternatives; that is, to take any set of alternatives and return a set whose sole element is their union. We therefore add this operator to the language, taking it to have no effect on a statements verifiers or falsifiers: $!|A|^+ = |A|^+$ and $!|A|^- = |A|^-$.

We define the notion of truth at a world in the usual way within truthmaker semantics (Fine 2017, 2021). We first designate a nonempty set $S^\diamond \subseteq S$ of possible states, subject to the constraint that whenever a state is possible, its parts are too (i.e. possibility is downward closed, if $s \in S^\diamond$ and $t \sqsubseteq s$ then $t \in S^\diamond$), and define a *world* to be a maximal possible state (that is, $w \in S$ is a world iff $w \in S^\diamond$ and $s \sqsubseteq w$ for all s such that $s \sqcup w \in S^\diamond$). We then define a statement A to be true at a world w iff w contains a verifier of an alternative of A ($s \in P$ for some $s \sqsubseteq w$ and $P \in \llbracket A \rrbracket$). We define as usual

²⁰One may adopt other frameworks in place of alternative semantics here; most notably, inquisitive semantics (Ciardelli, Groenendijk, and Roelofsen 2018). Inquisitive semantics differs from alternative semantics in that in the former, the semantic value of a statement consists of downward closed sets (where a set S is downward closed iff for all $x \in S$ and $y \sqsubseteq x$, $y \in S$). To recover the alternatives, inquisitive semantics takes the weakest propositions in the denotation: $alt(S) = \{p \in S : p \sqsubset q \text{ for no } q \in S\}$. This only makes a difference when one proposition in a denotation entails another, as with $A \vee (A \wedge B)$. Such Hurford-denotations are anyway odd (see Hurford 1974, Ciardelli and Roelofsen 2017, a.o.). Putting them aside, it does not matter here whether we use either alternative semantics, or inquisitive semantics with denotations filtered by *alt*.

²¹This is in the spirit of the generalised treatment of conjunction from Partee and Rooth (1983), adapted to the truthmaker setting: we define $a \sqcup_0 b = a \sqcup b$ and $a \sqcup_{i+1} b = \{c \sqcup_i d : c \in a, d \in b\}$, where $\sqcup_i$ is of type $\langle \sigma_i, \langle \sigma_i, \sigma_i \rangle \rangle$, σ_0 is the type of truthmaker states (s), and $\sigma_{i+1} = \langle \sigma_i, \mathbf{t} \rangle$.

that A entails B iff in every world where A is true, B is true.²²

Let $\llbracket A \rrbracket_M$ denote the semantic value of a statement A in model M according to alternative truthmaker semantics, defined as follows (as before, we omit the subscript for readability).

$$\begin{aligned}\llbracket p \rrbracket &= \{|p|^+\} \\ \llbracket !A \rrbracket &= \{|A|^+\} \\ \llbracket \neg A \rrbracket &= \{|\neg A|^+\} \\ \llbracket A \wedge B \wedge \dots \rrbracket &= \{P \sqcup Q \sqcup \dots : P \in \llbracket A \rrbracket, Q \in \llbracket B \rrbracket, \dots\} \\ \llbracket A \vee B \vee \dots \rrbracket &= \llbracket A \rrbracket \cup \llbracket B \rrbracket \cup \dots\end{aligned}$$

Some examples to illustrate the semantic clauses:

- (20) a. $\llbracket p \vee q \rrbracket = \{|p|^+, |q|^+\}$
 b. $\llbracket !(p \vee q) \rrbracket = \{|p|^+ \cup |q|^+\}$
 c. $\llbracket \neg(p \vee q) \rrbracket = \{|\neg(p \vee q)|^+\}$
 d. $\llbracket (p \vee q) \wedge r \rrbracket = \{|p|^+ \sqcup |r|^+, |q|^+ \sqcup |r|^+\}$

As the clauses show, the only connective capable of generating multiple alternatives is disjunction. The flattening operator and negation instead collapse alternatives: for any A , $\llbracket !A \rrbracket$ and $\llbracket \neg A \rrbracket$ always have a single element.

The resulting theory is a ‘two-storey’ semantics, in the sense of having two levels of semantic content. On the lower level sit the truthmakers for sentences that lack overt disjunction, such as ‘Adam picks more than four apples’ and ‘Eve eats infinitely many of the apples’. On the higher level sit the alternatives.

This theory achieves its goal of distinguishing both between the Eve pair in (3) and between the Adam pair in (21).

- (3) a. Eve eats infinitely many of the apples $a_1, a_2, a_3, \dots$
 b. Eve eats infinitely many of the apples $a_0, a_1, a_2, a_3, \dots$
- (21) a. Adam eats more than four pears.
 b. Adam eats five or more pears.

Concerning the first, since ‘Eve eats infinitely many of the apples $a_1, a_2, \dots$ ’ and ‘Eve eats infinitely many of the apples $a_0, a_1, a_2, \dots$ ’ do not contain overt disjunction, we assume that they have a single alternative. In the present framework, we achieve this

²²This corresponds to what Fine calls *loose* entailment, updated with alternatives.

result via the flattening operator, by parsing the sentences as $! \bigvee_{A \subseteq \infty \{a_1, a_2, \dots\}} \bigwedge_{a \in A} Pa$ and $! \bigvee_{A \subseteq \infty \{a_0, a_1, a_2, \dots\}} \bigwedge_{a \in A} Pa$, respectively. This returns:

- (22) a. $\llbracket \text{Eve eats infinitely many of the apples } a_1, a_2, \dots \rrbracket$
 $= \{ | \bigvee_{A \subseteq \infty \{a_1, a_2, \dots\}} \bigwedge_{a \in A} Pa |^+ \}$
 b. $\llbracket \text{Eve eats infinitely many of the apples } a_0, a_1, a_2, \dots \rrbracket$
 $= \{ | \bigvee_{A \subseteq \infty \{a_0, a_1, a_2, \dots\}} \bigwedge_{a \in A} Pa |^+ \}$

Each sentence has one alternative—the set of its truthmakers—and since they have different truthmakers, they correspondingly have different alternatives too.

Turning to the Adam pair, these sentences are not distinguished by their truthmakers but by their alternatives. We assume, plausibly, that (21a) and (21b) have same truthmakers (the state of Adam eating n pears, for $n = 5, 6, \dots$). Since the first lacks overt disjunction, it has a single alternative, while since the second has an overt disjunction, it has multiple. We assume that for each n , ‘Eve eats n pears’ and ‘Eve eats more than n pears’ are logically atomic, denoted by b_n and $b_{>n}$, respectively. Thus (21a) is parsed as $b_{>4}$ and (21b) as $b_5 \vee b_{>5}$. We therefore have:

- (23) a. $\llbracket \text{Adam eats more than four pears} \rrbracket = \{ | b_{>4} |^+ \}$
 b. $\llbracket \text{Adam eats five or more pears} \rrbracket = \{ | b_5 |^+, | b_{>5} |^+ \}$

The Eve pair and the Adam pair illustrate the need for both levels of semantic content: the truthmakers and the alternatives. The Eve pair have different truthmakers, while the Adam pair have the same truthmakers but different alternatives: $| b_{>4} |^+ = | b_5 \vee b_{>5} |^+$ but $\llbracket b_{>4} \rrbracket \neq \llbracket b_5 \vee b_{>5} \rrbracket$.

Alternative truthmaker semantics accounts for the contrast in (17) by invalidating De Morgan’s Law. $\neg(A \wedge B)$ have the same truthmakers but different alternatives: $| \neg(A \wedge B) |^+ = | \neg A \vee \neg B |^+$ but $\llbracket \neg(A \wedge B) \rrbracket \neq \llbracket \neg A \vee \neg B \rrbracket$. This is because negation flattens alternatives while disjunction introduces alternatives: $\llbracket \neg(p \wedge q) \rrbracket = \{ | p |^- \cup | q |^- \}$ has one alternative, while $\llbracket \neg p \vee \neg q \rrbracket = \{ | p |^-, | q |^- \}$ has two.

5.2 Adding permission

Now that we have a semantic framework that determines the semantic content of each statement, let us state what it means for a statement to be permitted. Our starting point is the treatment of permission within alternative semantics from Aloni (2007). Aloni proposes that a statement A is permitted is true just in case for every alternative p of A , there is a deontically accessible world where p is true. This theory builds on the possible worlds analysis of permission from standard deontic logic (Hanson 1965). It therefore

does not distinguish between the sentences in the Eve pair, (3), since the deontically accessible worlds where Eve eats infinitely many of $a_0, a_1, a_2, \dots$ are the same as those where she eats infinitely many of $a_1, a_2, \dots$.

To transpose Aloni’s clause into alternative truthmaker semantics, we will modify it in two ways. The first change is to replace the standard clause for permission (truth in some deontically accessible world) with Fine’s (2014) clause, that some verifier is permitted. The second change is to assume, following Goldstein (2019), that alternatives are subject to a homogeneity presupposition to the effect that the alternatives have the same status: whatever permission demands of one alternative, it demands of all. This all-or-none condition will ensure dual prohibition, that ‘Adam may not have ice cream or cake’ implies that she may not have ice cream and that she may not have cake.

Given these two changes, I propose the following truth conditions for permission statements, where A is in our language $\mathcal{L}$.

(24) **Alternative truthmaker semantics for permission**

- a. $\llbracket \Diamond A \rrbracket$ is defined iff $\forall P, Q \in \llbracket A \rrbracket : (\exists s \in P : s \in C) \text{ iff } (\exists s \in Q : s \in C)$;
- b. if defined, $\llbracket \Diamond A \rrbracket$ is true iff $\forall P \in \llbracket A \rrbracket \exists s \in P : s \in C$

This says that a permission statement $\Diamond A$ is defined iff all or none of A ’s alternatives contain a permitted state; and if defined, is true iff all of them do.²³

This clause guarantees free choice, that $\Diamond(A \vee B)$ implies $\Diamond A$ and $\Diamond B$. This captures the contrast in the Adam pair. Given that Adam may eat at most five pears, ‘Adam may eat more than four pears’ is true: $\llbracket b_4 \rrbracket = \{|b_{>4}|^+\}$ and some exact verifier of $b_{>4}$ is permitted; namely, eating exactly five pears. But ‘Adam may eat five or more pears’ is not true but rather undefined, since the alternatives differ in status: eating exactly five pears is permitted, while eating more than five is not.

Turning to the failure of De Morgan’s Law, where p denotes working Saturday and q working Sunday, $\llbracket \neg(p \wedge q) \rrbracket = \{|p|^- \cup |q|^- \}$ while $\llbracket \neg p \vee \neg q \rrbracket = \{|p|^-, |q|^- \}$. Hence one has permission to not work both days this weekend, $\Diamond \neg(p \wedge q)$, iff some falsifier of ‘work Saturday’ or ‘work Sunday’ is permitted. This is compatible with a requirement to work Saturday. But one has permission to take Saturday or Sunday off, $\Diamond(\neg p \vee \neg q)$, iff some falsifier of ‘work Saturday’ is permitted *and* some falsifier of ‘work Sunday’ is permitted. This is not compatible with a requirement to work Saturday, thereby deriving the contrast in (17).

²³Given the homogeneity presupposition and the fact that every sentence has some alternative (ensured by our semantic clauses), the use of universal quantification in (24b) is not essential: we may equivalently take $\llbracket \Diamond A \rrbracket$, if defined, to be true iff *some* alternative of A is permitted.

5.3 Embedding data

The present theory improves on Fine’s account regarding the embedding data from Section ???. The homogeneity assumption captures dual prohibition, that $\neg\Diamond(A \vee B)$ implies $\neg\Diamond A$ and $\neg\Diamond B$. Whenever $\neg\Diamond(A \vee B)$ is defined, both $\Diamond A$ and $\Diamond B$ will have the same truth value. Given $\neg\Diamond(A \vee B)$, they cannot both be true, so they must both be false: $\neg\Diamond A$ and $\neg\Diamond B$.

The theory has the flexibility to handle the ambiguity in probability judgements. Recall in the dessert scenario that ‘What is the probability that Alice may eat ice cream or cake?’ has two possible answers: two thirds and zero. The present theory predicts exactly these. It provides two options for what the question could be asking: for the probability that $!(p \vee q)$ is permitted, or that $p \vee q$ is. The first has a single alternative: $\llbracket !(p \vee q) \rrbracket = \{|p|^+ \cup |q|^+\}$. On this reading, Alice is permitted to have ice cream (p) or cake (q) iff some verifier of p or q is true, i.e. in worlds where the die lands 3–6, a chance of two-thirds. The second has two alternatives: $\llbracket (p \vee q) \rrbracket = \{|p|^+, |q|^+\}$. On this reading, Alice is permitted to have ice cream or cake iff some verifier of p is permitted *and* some verifier of q is permitted. But this is clearly false: no matter how the die lands, at most one dessert is permitted. It therefore has probability zero. In this way, the present theory provides for both probability judgements we observe.

5.4 Universal realisation inferences in alternative truthmaker semantics

Our clause in (24) gives the literal truth conditions of permission statements, but not their pragmatically derived, universal realisability inferences. As before, we take the universal realisation inference to strengthen the existential quantifier in the descriptive clause to universal: for each alternative, each of its verifiers is permitted.

(25) **Universal realisation inferences in alternative truthmaker semantics.**

The inference from $\Diamond A$ to $\forall p \in \llbracket A \rrbracket \forall s \in p : s \in C$

To illustrate, the universal realisation inference from ‘You are allowed to paint the wall green or blue’ will be that every shade of green *and* every shade of blue is permitted. This inference arises in contexts conducive to universal realisation inferences, such as in response to a question ‘What colours may I paint the wall?’ and on a prescriptive use of the permission statement.

5.5 Truthmaker semantics for permission

This gives us the truth conditions of permission statements. It does not, however, tell us what state verifies a permission statement. This would be required to give recursive clauses in full generality, to determine exact entailment between permission statements (where one statement exactly entails another iff every verifier for the first verifies the second), as well as to preserve the familiar idea that a statement is true (respectively, false) at a world iff the world contains an exact verifier (falsifier) of the statement. To do so, let us follow Fine (2018:644) in taking C to also be a state ($C \in S$); in effect, identifying the code of conduct with the state that it prevails.

We then define that this state C verifies $\Diamond A$ iff it is true according to C , and falsifies $\Diamond A$ iff it is false according to C . Since permission is not ‘disjunctive’, we take permission statements to have a single alternative. This gives us the following clauses.

$$\begin{aligned} |\Diamond A|^+ &= \begin{cases} \{C\} & \text{if } \forall P \in \llbracket A \rrbracket \exists s \in P : s \in C \\ \emptyset & \text{otherwise} \end{cases} \\ |\Diamond A|^- &= \begin{cases} \{C\} & \text{if } \forall P \in \llbracket A \rrbracket \forall s \in P : s \notin C \\ \emptyset & \text{otherwise} \end{cases} \\ \llbracket \Diamond A \rrbracket &= \{|\Diamond A|^+\} \end{aligned}$$

These automatically incorporate our homogeneity assumption. In cases where one alternative is permitted and another is not—as in ‘You may post the letter or burn down the post office’—the statement will have neither a verifier nor a falsifier. And on the standard assumption that a statement is true (respectively, false) at a world iff the world contains an exact verifier (falsifier) of it, at worlds that contain neither a verifier nor falsifier of the statement, it will be neither true nor false.

6 Independent motivation from conditionals

Beyond permission, there is independent motivation for alternative truthmaker semantics. The very same behaviour that motivates adopting the framework for permission also appears in conditional antecedents. Specifically, this section presents four parallels between permission and conditional antecedents that alternative truthmaker semantics is able to account for. They are:

1. Failures of Substitution;
2. Strong disjunction in both positive and negative environments;
3. Ambiguity between strong and weak disjunction with probability;

4. Failures of De Morgan's Law.

6.1 Failures of Substitution

Let Conditional Substitution be the principle that whenever two statements A and B are necessarily equivalent, then so are the conditionals $A > C$ and $B > C$, where $A > C$ denotes *if A, would C*. Fine (2014:328) observes that Alternative Eden also provides a counterexample to Conditional Substitution.

- (26) a. If Eve ate infinitely many of the apples $a_1, a_2, \dots$, she would be OK.
 b. If Eve ate infinitely many of the apples $a_0, a_1, a_2, \dots$, she would be OK.

Intuitively, the first is true but the second is not. Only the second implies that if Eve ate the Forbidden Fruit a_0 alongside infinitely many of the apples, she would be OK. The very same necessarily equivalent sentences that come apart under permission also come apart in conditional antecedents.

6.2 Strong disjunction in both positive and negative environments

Recall from Section 4.1 that $\Diamond(A \vee B)$ is strong in both positive and negative environments: $\Diamond(A \vee B)$ implies $\Diamond A \wedge \Diamond B$, and also, $\neg\Diamond(A \vee B)$ implies $\neg\Diamond A \wedge \neg\Diamond B$.

Conditionals behave in the same way.

- (27) a. All of my friends would have fun at the party if Alice or Bob went.
 b. None of my friends would have fun at the party if Alice or Bob went.

(27a) intuitively implies both that all of the friends would have fun if Alice went, and that all of them would have fun if Bob went—an instance of Simplification of Disjunctive Antecedents (SDA). Equally, Santorio (2018:524) observes that (27b) intuitively implies both that none of the friends would have fun if Alice went, and that none of them would have fun if Bob went.²⁴ In this case, $(A \vee B) > C$ implies $(A > C) \wedge (B > C)$ and $\neg(A \vee B) > C$ implies $\neg(A > C) \wedge \neg(B > C)$ —parallel to permission.

6.3 Ambiguity between strong and weak disjunction with probability

Recall from the dessert scenario in Section 4.2 that *The probability that you may A or B* has two readings: a strong, free choice reading and a weak, non-free choice reading.

²⁴Following the strategy in (18), I have phrased the examples using *none* to fix the negation's scope above the entire conditional as $\neg\exists x((Ax \vee Bx) > Cx)$.

Conditionals exhibit analogous behaviour, with SDA in place of free choice. *If A or B, probably C* is ambiguous between a strong, SDA-validating reading and a weak, SDA-invalidating reading.

To bring out SDA reading, consider the following case from Santorio (2021:633).²⁵

Alice threw a fair six-sided die. Before her throw, Bob bet that the outcome would be a 3 or a 4. Mary said that the die would land 2, 3, or 4; Sue claimed that it would land 3, 4, or 5. In fact it landed 1, so Bob lost.

Consider:

- (28) If Mary's report had been accurate or Sue's report had been accurate, Bob probably would have won.

This has a prominent true reading. The reasoning behind this true judgement is that, if Mary's had been accurate, the die would have landed 2, 3, 4, in which case is a two thirds chance of Bob winning; similarly if Sue's report had been accurate. Either way, then, there would be a more than half chance of Bob winning—what *probably* requires.

This is the SDA reading, on which *if A or B, probably C* is equivalent to the conjunction of *if A, probably C* and *if B, probably C*: we suppose each disjunct separately and check whether the consequent holds. Other cases bring out the non-SDA reading. Consider:

- (29) If the die had landed 2, 3, or 4, Bob probably would have won.

This also has a true reading. The reasoning behind this is that, in two of three the cases where the die lands 2, 3, or 4, Bob wins, which is greater than one half. This is the non-SDA reading. In contrast, the SDA-reading is not true. On this reading, (29) entails that if the die had landed 2, Bob probably would have won, which is false.

6.4 Failures of De Morgan's Law

In (17) we saw that permission breaks De Morgan's Law. Ciardelli, Zhang, and Champollion (2018) present experimental evidence that conditional antecedents also violate De Morgan's Law: $\neg(A \wedge B) > C$ is not equivalent to $(\neg A \vee \neg B) > C$. In their scenario, there are two switches, A and B, connected to a light. Each switch can be up or down. The light is on just in case the switches agree, i.e. are either both up or both down. Currently, both switches are up, so the light is on. Compare:

²⁵I have modified Santorio's case to use the subjunctive where Santorio uses the indicative. This is purely to maintain consistency with our discussion up to now, which has used the subjunctive.

- (30) a. If switch A or switch B was down, the light would be off. $(\bar{A} \vee \bar{B}) > C$
 b. If the switches were not both up, the light would be off. $\neg(A \wedge B) > C$

The antecedents form an instance of De Morgan’s Law on the natural assumption that *down* is equivalent to *not up* ($\bar{A} \equiv \neg A$ and $\bar{B} \equiv \neg B$). Ciardelli, Zhang, and Champollion found that a majority of participants (69%) judged (30a) to be true but only a small minority (22%) judged (30b) to be true.

6.5 An alternative truthmaker semantics of conditionals

Let us see how alternative truthmaker semantics can uniformly account for the parallel behaviour of permission and conditional antecedents. Recall that for permission, we took an condition on sets of states—the existence of a permitted states in the set—and added homogeneous quantification over alternatives. For conditionals, we will do exactly the same. We take an operation on sets of states—this time a conditional operator—and add the same alternative structure.

Where P and Q are sets of states, we assume a basic conditional operation $P \Rightarrow Q$. This may be analysed this in terms of existing state-based theories of conditionals, such as Fine (2012) or McHugh (2022, 2023).²⁶

Assuming an analysis of $\Rightarrow$, we then lift this basic conditional operation to the level of alternatives following the recipe laid out by Ciardelli (2016): a conditional $A > C$ is

²⁶For concreteness, let us briefly show how this may be done on these theories. Fine (2012) introduces a primitive transition relation $\rightarrow$ between states, whereby $s \rightarrow t$ represents that t is a ‘possible outcome’ of s (Fine takes the possible outcome relation to depend on the world of evaluation, which we omit here for ease of presentation). He takes a counterfactual $A > C$ to be true iff every verifier of A has a possible outcome that contains a verifier of C ($\forall s \in |A|^+ \exists t : s \rightarrow t \wedge \exists u \in |C|^+ : u \sqsubseteq t$). From this we may readily extract the conditional operation over arbitrary sets of states by replacing P with A ’s verifiers and Q with C ’s verifiers. This returns: $P \Rightarrow Q$ iff every state in P has a possible outcome that contains a state in Q , i.e. $\forall s \in P \exists t (s \rightarrow t \wedge \exists u \in Q : u \sqsubseteq t)$.

McHugh (2022, 2023) introduces an *aboutness* relation between sentences and states, determining what parts of the world a sentence is about. Where P is a set of states, He formalises what it means to ‘vary’ the states in P , and defines the notion a world being a ‘ P -variant’, meaning that it results from varying the states in P . He further defines what it means for one world to result from playing the laws forward at a world, and takes a counterfactual $A > C$ to be true at w just in case C is true at every world that results from playing the laws forward from an P -variant of w , where P is the set of states A is about. From this we may also easily extract a conditional operation over states: $P \Rightarrow Q$ iff every world that results from playing the laws forward from some P -variant contains a state in Q .

For present purposes, it does not matter which of these analyses of $\Rightarrow$ we plug in. What matters is that they both invalidate Conditional Substitution. For Fine, this is because necessarily equivalent sentences can have different verifiers. For McHugh, it is because necessarily equivalent sentences can be about different things.

true iff each of alternative of A implies an alternative of C . To Ciardelli's condition we add homogeneous quantification over alternatives, mirroring our entry for permission in (24).

(31) **Alternative truthmaker semantics for conditionals**

- a. $\llbracket A > C \rrbracket$ is defined iff
 $\forall P, Q \in \llbracket A \rrbracket : (\exists R \in \llbracket C \rrbracket : P \Rightarrow R) \text{ iff } (\exists R \in \llbracket C \rrbracket : Q \Rightarrow R);$
- b. if defined, $\llbracket A > C \rrbracket$ is true iff $\forall P \in \llbracket A \rrbracket \exists R \in \llbracket C \rrbracket : P \Rightarrow R$.

This clause accounts for the behaviour of conditional antecedents above.

First, we account for the Eve pair in (26). Defining $\Rightarrow$ over sets of truthmakers allows Substitution to fail, even for sentences like those in the Eve pair with a single alternative.²⁷

Second, the homogeneity assumption ensures that conditionals with disjunctive antecedents are strong in positive and negative environments.

Third, we may account for the ambiguity between strong and weak disjunction in conditionals antecedents just as we did for permission in Section 5.3, by taking disjunction under probability to admit both the unflattened reading $A \vee B$ and the flattened reading $!(A \vee B)$. Only the unflattened reading validates SDA: $(A \vee B) > C$ entails $(A > C) \wedge (B > C)$, but $!(A \vee B) > C$ does not.

Finally, we account for Failures of De Morgan's Law. $(\bar{A} \vee \bar{B}) > C$ is equivalent to $(\bar{A} > C) \wedge (\bar{B} > C)$. Since each conjunct is true, we predict (30a) to be true. However, the status of (30b) depends on the underlying conditional $\Rightarrow$. If $|A|^- \cup |B|^- \Rightarrow \text{OFF}$ fails, (30b) is false.

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²⁷Where P_0 be the set of verifiers of 'Eve eats infinitely many of $a_0, a_1, a_2, \dots$ ', P_1 the set of verifiers for 'Eve eats infinitely many of $a_1, a_2, \dots$ ', and OK the set of verifiers for 'Eve is OK', one may show that both Fine and McHugh's theories predict $P_0 \Rightarrow OK$ to be false and $P_1 \Rightarrow OK$ to be true on the natural representations of Alternative Eden in their theories; for details see Fine (2012) and McHugh (2023:157).

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